DISCRETE STRUCTURES

CMSC250

Chapter 13: Sequences, Summations, and Products

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Indexed patterns, sums, products, and recurrences



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1 Sequences: values indexed by integers

A sequence is an ordered list of terms, typically written $a_0, a_1, a_2, \ldots$ or $a_1, a_2, \ldots$. Its index is part of the definition: the sequence $a_n = 2n + 1$ begins $1, 3, 5, \ldots$ when $n = 0$, but begins $3, 5, 7, \ldots$ when $n = 1$.

Key idea. Always identify the first index. A correct formula with the wrong starting index gives the wrong sum or term.

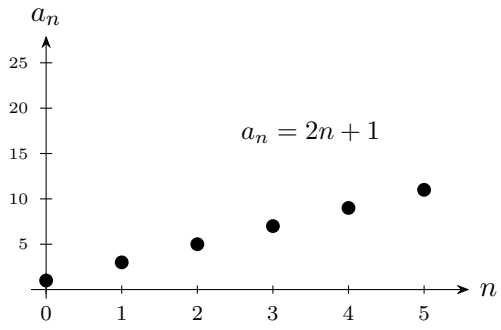
Worked example: evaluate an explicitly defined sequence

Question. Let $a_n = 3n^2 - 2n$ for $n \geq 0$. Find a_0, a_1, a_2, a_3.

Solution. Substitute the indices: $a_0 = 0$, $a_1 = 3 - 2 = 1$, $a_2 = 12 - 4 = 8$, and $a_3 = 27 - 6 = 21$. The first four terms are $0, 1, 8, 21$.

2 Visualizing a sequence

A sequence can be plotted as separate points (n, a_n); unlike a continuous function graph, there are no terms at fractional indices.



3 Arithmetic and geometric sequences

An **arithmetic sequence** adds a fixed difference d: $a_n = a_0 + nd$ (for $n \geq 0$). Example: $5, 8, 11, 14, \ldots$ has $a_n = 5 + 3n$. A **geometric sequence** multiplies by a fixed ratio r: $a_n = a_0 r^n$. Example: $2, 6, 18, 54, \ldots$ has $a_n = 2 \cdot 3^n$.

Worked example: identify and predict terms

Question. Find a formula for $7, 12, 17, 22, \ldots$ beginning at $a_1 = 7$.

Solution. The common difference is 5, so $a_n = 7 + 5(n - 1) = 5n + 2$ for $n \geq 1$. In particular $a_{10} = 52$.

4 Sigma notation: reading the bounds

The notation $\sum_{k=p}^{q} f(k)$ means add $f(k)$ for every integer k from p through q inclusive:

$$\sum_{k=2}^{5} k^2 = 2^2 + 3^2 + 4^2 + 5^2 = 4 + 9 + 16 + 25 = 54.$$

The symbol k is a *dummy index*; replacing it consistently with j does not change the sum.

$$\boxed{\underbrace{4 + 9 + 16 + 25}_{k=2,3,4,5} = 54}$$

5 Basic sum rules

For constants c and functions f, g,

$$\sum_{k=p}^{q} (cf(k) + g(k)) = c \sum_{k=p}^{q} f(k) + \sum_{k=p}^{q} g(k).$$

You may split a sum into consecutive index ranges:

$$\sum_{k=1}^{n} f(k) = \sum_{k=1}^{r} f(k) + \sum_{k=r+1}^{n} f(k) \quad (1 \le r < n).$$

A constant summed n times is $\sum_{k=1}^{n} c = cn$. Be careful that $\sum_{k=0}^{n} c = c(n+1)$ because there are $n+1$ terms.

6 Three formulas to know and understand

For $n \ge 1$,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Worked example: sum consecutive integers by pairing

Question. Explain why $1 + 2 + \cdots + n = n(n+1)/2$.

Solution. Write $S = 1 + 2 + \cdots + n$ and reverse the order: $S = n + (n-1) + \cdots + 1$. Adding term by term gives $2S = (n+1) + (n+1) + \cdots + (n+1)$ with n copies, so $2S = n(n+1)$ and $S = n(n+1)/2$.

Worked example: evaluate a polynomial sum

Question. Compute $\sum_{k=1}^{10} (3k^2 - 2k + 4)$.

Solution. Split the sum:

$$\begin{aligned}\sum_{k=1}^{10} (3k^2 - 2k + 4) &= 3\sum_{k=1}^{10} k^2 - 2\sum_{k=1}^{10} k + \sum_{k=1}^{10} 4 \\ &= 3(385) - 2(55) + 40 = \boxed{1085}.\end{aligned}$$

7 Geometric sums and cancellation

For $r \neq 1$,

$$\boxed{\sum_{k=0}^{n} r^k = \frac{r^{n+1}-1}{r-1} = \frac{1-r^{n+1}}{1-r}.}$$

If $r = 1$, there are $n+1$ terms, so the sum is $n+1$.

Worked example: derive the geometric-sum formula

Question. Find $S = 1 + r + r^2 + \cdots + r^n$ for $r \neq 1$.

Solution. Multiply by r: $rS = r + r^2 + \cdots + r^{n+1}$. Subtract the two equations: $S - rS = 1 - r^{n+1}$. Thus $(1-r)S = 1 - r^{n+1}$. Because $r \neq 1$, divide to obtain $S = (1-r^{n+1})/(1-r)$.

Worked example: evaluate a finite geometric sum

Question. Compute $\sum_{k=0}^{5} 2^k$.

Solution. Apply the formula with $r = 2$ and $n = 5$: $(2^6-1)/(2-1) = 63$. Directly: $1+2+4+8+16+32 = 63$.

8 Changing the index without changing the sum

Reindexing is useful when expressions start at awkward bounds. Set $j = k - 2$. As k runs from 2 to 5, j runs from 0 to 3. Thus

$$\sum_{k=2}^{5} (k-2)^2 = \sum_{j=0}^{3} j^2 = 0 + 1 + 4 + 9 = 14.$$

Key idea. When changing indices, transform *both* the summand and the lower/upper bounds. Changing only the variable name is not reindexing.

Worked example: a shifted arithmetic sum

Question. Compute $\sum_{k=4}^{9} (k-3)$.

Solution. Let $j = k - 3$. When $k = 4$, $j = 1$; when $k = 9$, $j = 6$. Thus $\sum_{k=4}^{9}(k-3) = \sum_{j=1}^{6} j = 6 \cdot 7/2 = 21$.

9 Telescoping sums

A telescoping sum has terms that cancel with neighboring terms.

Worked example: a rational telescoping sum

Question. Evaluate $\sum_{k=1}^{n} \frac{1}{k(k+1)}$.

Solution. First use partial fractions:

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Therefore

$$\begin{aligned}\sum_{k=1}^{n} \frac{1}{k(k+1)} &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} = \boxed{\frac{n}{n+1}}.\end{aligned}$$

Only the first positive term and last negative term survive.

10 Product notation

The product symbol $\prod$ is the multiplicative counterpart of $\sum$:

$$\prod_{k=1}^{4} k = 1 \cdot 2 \cdot 3 \cdot 4 = 24 = 4!, \qquad \prod_{k=1}^{n} k = n!.$$

By convention, the empty product is 1 and the empty sum is 0.

Worked example: a telescoping product

Question. Evaluate $\prod_{k=1}^{n} \frac{k+1}{k}$.

Solution. Expand and cancel:

$$\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \boxed{n+1}.$$

11 Recursive sequences and initial conditions

An explicit formula directly calculates a_n from n. A **recurrence** defines a term using earlier terms and requires initial conditions.

Worked example: trace a recurrence

Question. Given $a_0 = 2$ and $a_n = 3a_{n-1} + 1$ for $n \geq 1$, find a_1, a_2, a_3.

Solution. $a_1 = 3(2) + 1 = 7$, $a_2 = 3(7) + 1 = 22$, and $a_3 = 3(22) + 1 = 67$. Every step depends on the preceding term.

Worked example: two previous terms

Question. Given $a_0 = 1$, $a_1 = 2$, and $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 2$, find a_2, a_3, a_4.

Solution. $a_2 = 2 + 2(1) = 4$, $a_3 = 4 + 2(2) = 8$, $a_4 = 8 + 2(4) = 16$. Two initial conditions are necessary because each new term uses the two preceding terms.

12 How to prove a sum identity

An identity with a variable upper bound is a statement about *all* positive integers, not merely the first few. Direct algebra can prove certain formulas (such as the pairing and telescoping arguments above). Mathematical induction provides another general method, developed in its own chapter. Numerical checking is useful for catching errors, but cannot replace a universal proof.

13 Common mistakes

- Starting a sum at $k = 0$ but using a formula that assumes $k = 1$.
- Counting n terms in $\sum_{k=0}^{n}$ instead of $n + 1$.
- Using the geometric formula without separately handling $r = 1$.
- Changing the summation index but not its bounds.
- Mistaking a recursive definition for a closed-form formula.
- Cancelling factors across sums: cancellation is valid in telescoping expressions only after algebraic expansion.

14 Practice questions

1. Find the first five terms of $a_n = (-1)^n(n + 1)$, starting at $n = 0$.
2. Compute $\sum_{k=3}^{8}(2k - 1)$.
3. Compute $\sum_{k=0}^{6} 3^k$.
4. Evaluate $\sum_{k=1}^{n} \frac{1}{k(k+2)}$.
5. Evaluate $\prod_{k=2}^{n} \frac{k}{k-1}$ for $n \geq 2$.
6. Given $a_0 = 1$, $a_1 = 3$, and $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$, find a_2, a_3, a_4.

15 Fully worked practice solutions

1. Substitute $n = 0, 1, 2, 3, 4$: $1, -2, 3, -4, 5$.
2. $\sum_{k=3}^{8}(2k - 1) = 2\sum_{k=3}^{8} k - 6 = 2(3 + 4 + 5 + 6 + 7 + 8) - 6 = 2(33) - 6 = 60$.

3. By the geometric formula, $\sum_{k=0}^{6} 3^k = (3^7 - 1)/(3 - 1) = (2187 - 1)/2 = 1093$.

4. Use $\frac{1}{k(k+2)} = \frac{1}{2}(\frac{1}{k} - \frac{1}{k+2})$. The sum telescopes, leaving

$$\frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right) = \boxed{\frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}}.$$

5. Expand: $(2/1)(3/2)(4/3)\cdots(n/(n-1)) = n$.

6. $a_2 = 2(3) - 1 = 5$, $a_3 = 2(5) - 3 = 7$, and $a_4 = 2(7) - 5 = 9$.